

## Review

# Genomic privacy and security in the era of artificial intelligence and quantum computing

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© The Author(s) 2025 **OPEN****Abstract**

The rapid advancements in sequencing technologies have greatly increased access to genomic data stored in public databases. This has raised significant privacy and security concerns. This review emphasizes the importance of protecting genomic data by analyzing vulnerabilities in current storage and sharing practices. It examines the risks genetic databases face from cyber-attacks and internal breaches, focusing especially on advanced AI-driven threats and quantum computing vulnerabilities. The review explores machine learning methods designed to secure data. It highlights algorithms that prioritize privacy while maintaining data confidentiality, such as differential privacy, federated learning, and synthetic data generation using Generative Adversarial Networks (GANs). Findings demonstrate progress in mitigating common privacy breaches like re-identification and inference attacks. However, persistent vulnerabilities remain, particularly to emerging threats such as model inversion and membership inference attacks. The review advocates an integrated approach combining robust legislative frameworks with advanced technology to address genomic privacy challenges. It calls for intensified research efforts to safeguard genomic information. In particular, there is an urgent need to adopt quantum-resistant cryptographic methods, including lattice-based encryption and blockchain-integrated security frameworks. The paper emphasizes the necessity for genomics researchers to prioritize data privacy and security. This ensures responsible handling of genomic information in research.

**Keywords** Genomic data · Cyber-security · Privacy · Machine learning · Quantum-resistant cryptography

## 1 Introduction

Advancements in genome sequencing and genetic analysis technologies have significantly reduced the cost of sequencing and analyzing DNA. This has increased the accessibility and sharing of genomic data across various fields. In biotechnology, genomic data provides detailed genetic insights. In forensic sciences, genetic data is used extensively in criminal investigations. However, the widespread sharing of genomic data introduces substantial privacy and security concerns. Malicious individuals can exploit this data to identify individuals, leading to potential privacy breaches, identity tracing, extortion, or unauthorized surveillance [1].

Companies and research organizations often store genomic data in third-party cloud services. This increases the vulnerability of genomic data to cyber threats and unauthorized access. Such practices expose genomic datasets to re-identification, membership inference, and linkage attacks, often facilitated by public databases and social media

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platforms [2]. Additionally, genomic data analysts frequently prioritize speed and accuracy over security. This approach increases the susceptibility of genomic software to vulnerabilities and cyber threats.

This review addresses the crucial balance between data utility and privacy. It systematically identifies key vulnerabilities and evaluates privacy-preserving techniques used in genomic data [2–4]. We discuss various technical solutions including data perturbation, cryptographic methods, and AI-driven defenses. The review also explores legal and ethical considerations to offer a comprehensive understanding of genomic privacy risks and protective measures. By reviewing current research and policy frameworks, we highlight existing gaps and suggest future research directions to improve genomic data security.

We categorize literature related to genomic privacy and security. The focus is on technical vulnerabilities, privacy-preserving techniques, and regulatory frameworks. This review follows a **structured literature review (SLR)** approach. It includes peer-reviewed journal articles, conference papers, and regulatory guidelines. The goal is to ensure a comprehensive and up-to-date analysis. Our methodology involves selecting, analyzing, and evaluating relevant academic studies and policy documents. We apply **inclusion and exclusion criteria** to refine our selection process:

- **Inclusion criteria:** We include studies on novel security techniques, AI-driven privacy risks, cryptographic protections, and recent regulatory changes. We also prioritize research on comparative security frameworks, computational efficiency, and real-world applications.
- **Exclusion criteria:** We exclude studies that lack technical depth or focus only on ethical discussions without security solutions. We also remove outdated papers that do not cover recent AI-based genomic privacy risks. Studies that discuss general data privacy laws without relevance to genomics are not included.

We gather literature from *IEEE Xplore*, *ACM Digital Library*, and *Google Scholar*, *ArXiv*. We use a thematic categorization approach to structure the findings. The review is divided into genomic data attack types, cryptographic protections, AI-driven threats, and regulatory frameworks. The goal is to highlight key trends, research gaps, and future directions. To maintain relevance, we focus on papers published between 2020 and 2025.

The paper is structured as follows: Section 2 discusses genomic data attacks; Section 3 reviews privacy-preserving methods; and Section 4 covers regulatory and ethical frameworks. Later sections critically evaluate computational efficiency, AI-driven threats, and quantum-resistant solutions. The paper concludes by offering recommendations for future research directions.

## 2 Genomics data attacks

In this section, we categorize and explain various attacks on genomic data. We detail how these attacks are performed by adversaries. Genomic data is vulnerable to various types of privacy attacks. These attacks exploit statistical, computational, and AI-driven methods to extract sensitive genetic information. Table 1 summarizes these attacks.

Identity tracing attacks use quasi-identifiers such as age, sex, and location to re-identify individuals from de-identified genomic datasets. Techniques like genealogical triangulation and public record linkage enable adversaries to match genetic data with existing demographic information. This creates a significant privacy risk [5–8]. Also, inference attacks, including Attribute Disclosure and Membership Inference, allow attackers to infer sensitive phenotypic traits or determine an individual's participation in a genomic study. They do this by analyzing allele frequencies or dataset summary statistics. Genome-Wide Association Studies (GWAS) have been particularly susceptible to these attacks. Because of this, stricter access control measures are necessary [9–12]. Completion attacks, also known as genome reconstruction attacks, leverage genetic similarities among family members to infer missing genotypic information. High-profile cases, such as the Golden State Killer investigation, show how forensic genealogy and public DNA databases can be exploited for identity reconstruction [2, 13, 14].

Furthermore, advancements in machine learning and Bayesian networks have introduced novel threats to genomic data security. Bayesian networks have been used to predict missing genomic markers and perform identity inference. This remains possible even when differential privacy protections are applied [15, 16]. AI-driven machine learning-based re-identification techniques weaken de-identification strategies. They use deep learning models to match anonymized genomic sequences to identified datasets [17, 18]. Additionally, adversarial attacks on genomic AI models manipulate input data to mislead predictive models. This compromises their accuracy in genomic classification tasks [19–21]. These attacks include white-box

**Table 1** Summary of genomic privacy attacks and their mechanisms

| Attack type                              | Description                                                                                                                                              | Methods used                                                                                                                                                                                                                                                                                                                                                                                   | Notable examples                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identity Tracing Attacks                 | Adversaries use de-identified genomic data along with quasi-identifiers (e.g., age, sex, last name) to re-identify individuals.                          | <ul style="list-style-type: none"> <li>• Genealogical triangulation</li> <li>• Public record linkage</li> </ul>                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• <b>Re-identification via demographics:</b> Using date of birth, sex, and zip code to uniquely identify individuals [5–7].</li> <li>• <b>Re-identification using microbiome sequencing:</b> Likelihood scores applied to human genome reads to predict genetic sex and ancestry, achieving high re-identification accuracy [8].</li> </ul>                                 |
| Inference Attacks                        | Attackers use partial information to infer additional related data, potentially exposing sensitive genomic details.                                      | <ul style="list-style-type: none"> <li>• Attribute Disclosure Attacks (ADA): Infers individual traits from genomic datasets.</li> <li>• Membership Inference Attacks (MIA): Determines whether an individual's data is part of a dataset.</li> <li>• Statistical inference using allele frequencies: Exploits allele frequency variations to infer disease risks or genetic traits.</li> </ul> | <ul style="list-style-type: none"> <li>• <b>Attribute Disclosure Attack in GWAS:</b> Led to phenotype inference from allele frequencies [9].</li> <li>• <b>Membership Inference Attack on GWAS:</b> Demonstrated vulnerability in summary statistics [10].</li> <li>• <b>Enhanced inference attacks using allele frequency statistics:</b> Showed greater accuracy in identifying individuals [11, 12].</li> </ul> |
| Completion Attacks                       | Adversaries reconstruct all or part of an individual's genetic information using publicly available DNA data or a family member's DNA sequence.          | <ul style="list-style-type: none"> <li>• Genotype imputation: Filling in missing genetic information using statistical methods.</li> <li>• Familial reconstruction: Using relatives' DNA to predict an individual's genetic makeup.</li> <li>• Matching DNA profiles against known family trees</li> </ul>                                                                                     | <ul style="list-style-type: none"> <li>• <b>Golden State Killer case:</b> FBI identified Joseph DeAngelo using forensic genealogy via GEDMatch [2, 13].</li> <li>• <b>Haplotype identification via genetic correlation:</b> Identified unknown individuals by matching identical DNA regions across relatives [14].</li> </ul>                                                                                     |
| Bayesian Networks in Genetic Attacks     | Bayesian networks (BNs) are probabilistic graphical models that adversaries use to infer missing genomic data and relationships between genetic markers. | <ul style="list-style-type: none"> <li>• Bayesian inference models: Uses probability-based statistical modeling to predict missing genetic data.</li> </ul>                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>• <b>GWAS data reconstruction:</b> Bayesian methods have been used to reveal disease-associated genetic markers [15].</li> <li>• <b>Inference attacks using Bayesian networks:</b> Demonstrated re-identification of individuals from GWAS statistics, even when differential privacy protections were applied [16].</li> </ul>                                             |
| Machine Learning-Based Re-Identification | Exploits AI models trained on genomic sequences to match anonymized data with identified datasets, undermining de-identification efforts.                | <ul style="list-style-type: none"> <li>• Deep learning models for genomic sequence matching.</li> <li>• Clustering and feature extraction.</li> <li>• Adversarial learning in genomic data privacy.</li> </ul>                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Genome reconstruction attack in genomic data-sharing beacons using clustering and machine learning [17].</li> <li>• AI-based classification of genomic variants for privacy threats [18].</li> </ul>                                                                                                                                                                      |

Table 1 (continued)

| Attack type                              | Description                                                                                                                                                                                | Methods used                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Notable examples                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adversarial Attacks on Genomic AI Models | These attacks involve the deliberate manipulation of genomic AI models through adversarial perturbations, which can lead to misclassification or the extraction of sensitive genomic data. | <ul style="list-style-type: none"><li>• White-box attacks: Direct gradient-based adversarial examples designed to deceive AI-based genomic models.</li><li>• Black-box attacks: Adversarial perturbations without direct access to model gradients.</li><li>• Data poisoning attacks: Injecting adversarial samples to corrupt model training.</li><li>• Feature importance adversarial attacks: Targeting AI feature selection mechanisms to misguide prediction outcomes.</li></ul> | <ul style="list-style-type: none"><li>• <b>Misclassification of genomic variants:</b> Adversarial mutations disrupt population clustering and ancestry inference [19].</li><li>• <b>Data integrity compromise:</b> Poisoned genomic datasets degrade AI model performance and accuracy [20].</li><li>• <b>Privacy leakage:</b> Adversarial methods bypass AI security measures, revealing sensitive genomic features [21].</li></ul> |

adversarial perturbations, black-box attacks, and data poisoning. All of these methods have been shown to degrade model performance significantly.

As AI and cryptographic solutions evolve, a hybrid security approach is essential. It must integrate robust encryption, adversarially trained AI models, and privacy-preserving federated learning frameworks. These measures are necessary to safeguard genomic data against emerging threats.

### 3 Technical solutions to mitigate genetic attacks

Companies and laboratories are increasingly outsourcing genetic information to cloud services due to benefits such as computational resources and cost savings [22]. However, storing data on clouds makes it vulnerable to cyber-attacks, necessitating strict access controls to ensure only authorized individuals can access the data. In addition, genomic technologies like sequencers, now capable of sequencing a whole genome in days, are often connected Internet of Things (IoT) devices. These devices are vulnerable to security threats due to their full internet connectivity and inherent limitations in processing, memory, and power. To mitigate genetic data attacks, several techniques are employed as summarized in Table 2. These include data perturbation methods like differential privacy [23], access control, segmentation, and cryptographic techniques such as homomorphic encryption, secure hardware, and secure multiparty computation [24, 25]. Cryptographic tools protect data confidentiality from unauthorized access, while access control and data perturbation ensure data anonymity against privacy intrusions [2].

#### 3.1 Data perturbation methods

To protect privacy when sharing de-identified data, perturbation methods such as data transformation, aggregation, obfuscation, and synthetic data generation can be used. Data transformation modifies the format or structure of data. A common method is *k*-anonymity, which ensures each quasi-identifier sequence (e.g., race, birth) appears at least *k* times in a dataset [26]. For example, ZIP codes can be generalized to larger regions by reducing a 5-digit ZIP code (e.g., 02139) to a 3-digit prefix (e.g., 021xx). Generalizing ZIP codes in this way groups individuals into broader geographic categories. This makes it more likely that each group contains at least *k* people who share the same quasi-identifiers (like birth date, gender, and the generalized ZIP code), which satisfies the privacy condition of *k*-anonymity. As a result, it becomes harder for an adversary to isolate and identify individuals based on detailed location information. This is particularly important because 5-digit ZIP codes, when combined with birth date and gender, can uniquely identify over 50% of U.S. residents [27]. This type of generalization is a common technique to protect against identity tracing attacks that exploit quasi-identifiers.

Data aggregation summarizes data instead of revealing individual records. The Beacon Network, developed by the Global Alliance for Genomics and Health (GA4GH), preserves privacy by responding to user queries with a simple “yes” or “no” as to whether a specific allele exists in a dataset, without exposing raw data [2]. This binary response format restricts direct access to individual-level information. In addition, privacy is further supported by methods such as query aggregation, rate limiting, and the potential integration of differential privacy, which together reduce the risk of re-identification through repeated or cross-referenced queries [2]. Data obfuscation adds noise to mask sensitive information. A key method is differential privacy, which limits membership inference attacks through a privacy budget. While effective, it must balance privacy and data utility. Combining differential privacy with cryptographic methods or using a weaker adversarial model can improve data usability [2]. An algorithm satisfies differential privacy if its output remains nearly the same with or without an individual’s data [28]. The privacy parameter  $\epsilon$  (or  $\epsilon$ ) controls privacy strength; lower values add more noise, enhancing privacy [28, 29]. Equation (1) defines the conditions for a mechanism to meet differential privacy.

$$\frac{Pr[F(x) = S]}{Pr[F(x') = S]} \leq e^\epsilon \quad (1)$$

Equation (1) shows two neighboring datasets  $x$  and  $x'$  where they differ by one record. A mechanism  $F$  takes the neighboring datasets as inputs and the equation shows the probability ratio of  $F(x)=S$  to  $F(x')=S$  is at most  $e$  raised to the privacy budget  $\epsilon$ .

**Laplace Mechanism:** The Laplace mechanism addresses the challenge of adding noise to datasets by sampling noise from a Laplace distribution and using it to perturb each coordinate of the output. This noise is scaled by the sensitivity of the function, which measures the effect of a change in the dataset on the query result. Equation (2) defines

the sensitivity  $S(Q)$ , where  $D_1$  and  $D_2$  are neighboring databases, and  $Q$  is the query.  $\|Q(D_1) - Q(D_2)\|_1$  is the  $L_1$  distance between  $Q(D_1)$  and  $Q(D_2)$ , capturing the maximum change a single individual's data can cause in the function [30].

$$S(Q) = \max_{D_1, D_2} \|Q(D_1) - Q(D_2)\|_1 \quad (2)$$

The Laplace mechanism ensures  $\epsilon$ -differential privacy by injecting noise into each query result value. This noise follows an independently and identically distributed (i.i.d.) Laplace distribution with a mean of zero and scale  $\frac{S(Q)}{\epsilon}$ . The probability density function (pdf) for the Laplace distribution is given by Equation (3).

$$pdf(\eta) = \frac{\epsilon}{2S(Q)} \exp\left(-|\eta| \cdot \frac{\epsilon}{S(Q)}\right) \quad (3)$$

Figure 1 graphically represents the pdf of the Laplace distribution.

For example, to use the Laplace mechanism to add noise to the average of  $m$  numbers, given a set  $S$  of  $m$  numbers in the range  $[0, b]$ , the function  $f(S) = \frac{1}{|S|} \sum_{z \in S} z$  calculates the average. The mechanism  $M(S)$  adds noise  $\eta$  to  $f(S)$ :

$$M(S) = f(S) + \eta \quad (4)$$

The noise  $\eta$  follows a Laplace distribution with scale  $\frac{b}{m\epsilon}$ :

$$\eta \sim \text{Lap}\left(\frac{b}{m\epsilon}\right) \quad (5)$$

With high probability,  $M(S)$  approximates  $f(S)$  closely, achieving high accuracy with few errors. Equation (6) shows this probability for the error parameter  $\alpha$ :

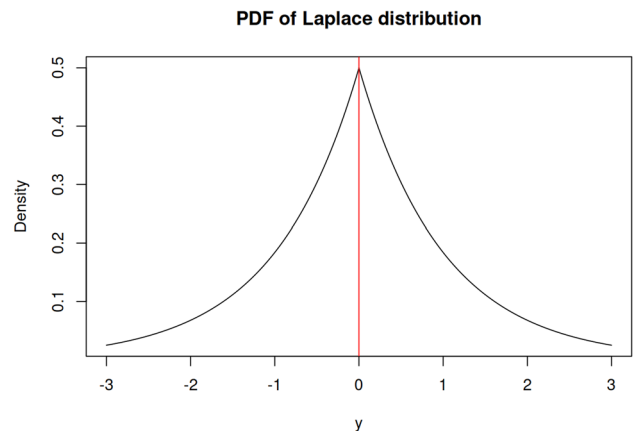
$$\begin{aligned} \Pr\left[\left|M(S) - \frac{1}{|S|} \sum_{z \in S} z\right| \geq \alpha\right] &= \Pr_{x \sim \text{Lap}\left(\frac{b}{m\epsilon}\right)}[|x| \geq \alpha] \\ &= \exp\left(-\frac{\alpha \epsilon m}{b}\right) \leq \beta \end{aligned} \quad (6)$$

Here,  $\alpha$  represents the error parameter, and  $\beta$  is the confidence parameter, showing the high probability of achieving accurate results.

**Exponential Mechanism:** The exponential mechanism is used for non-numeric query outputs, particularly when selecting an item from a finite set where noise addition is impractical. For example, choosing the best meeting day to maximize attendance. It uses a scoring function, the sensitivity of the scoring function, a dataset, possible outputs, and the privacy budget  $\epsilon$ . The scoring function assigns scores to all outputs, and the mechanism selects an output that is approximately the best, ensuring differential privacy.

A mechanism satisfies the exponential mechanism  $M_{u,\epsilon} : X \rightarrow Y$  with quality score  $u$  and parameter  $\epsilon$  if the outcome  $y$  is sampled with probability proportional to  $e^{\epsilon u(x,y)}$ .

**Fig. 1** Probability density function of the Laplace distribution



$$P[M_{u,e}(x) = y] = \frac{e^{\epsilon u(x,y)}}{\sum_{y \in Y} e^{\epsilon u(x,y)}} \quad (7)$$

Exponential mechanisms provide strong utility guarantees by quickly omitting lower-quality outcomes [31]. Dwork and Roth showed that the exponential mechanism ensures high utility by not significantly lowering the utility score of any returned element compared to the maximum score in the database [31].

DP mechanisms are proposed for genomics studies to provide privacy guarantees [32], but they do not consider data dependencies [33]. Data may be dependent due to social, behavioral, and genomic interactions [34, 35]. Adversaries could infer susceptibility to diseases using family genome correlations [36]. Researchers have shown that user dependencies can predict sensitive information from differentially private queries [34, 37].

## 3.2 Current machine learning methods for protecting genetic privacy and security

Machine learning has seen significant success in recent years and is now being applied in various fields, including bioinformatics, medicine, and health care. Research has focused on using machine learning models to protect sensitive information from leakage.

### 3.2.1 Differentiable private stochastic gradient descent (DP-SGD)

Zhang et al. [38] proposed the functional mechanism, a differential privacy technique designed for optimization-based analyses. This mechanism enforces  $\epsilon$ -differential privacy by perturbing the objective function of the optimization problem rather than its results. A randomized mechanism  $M$  satisfies  $\epsilon$ -differential privacy if, for any output from  $M$  and given two neighboring databases  $D_1$  and  $D_2$ , we have:

$$Pr[M(D_1) = O] \leq e^\epsilon \cdot Pr[M(D_2) = O] \quad (8)$$

They constructed an algorithm for regression analysis based on the Laplace mechanism, which samples noise from an independent and identically distributed Laplace distribution. The main challenge was perturbing the objective function in a differentially private manner, given its mathematical complexity and high dimensionality. Objective functions in regression often involve non-linear relationships and multiple variables, making it difficult to directly apply noise without degrading model performance. Zhang et al. in [38] addressed this by using the polynomial representation of the objective function, injecting Laplace noise into its polynomial coefficients, and then deriving the model parameter that minimizes the perturbed function. Their algorithm satisfied  $\epsilon$ -differential privacy, and extensive experiments showed it achieved highly accurate regression results, outperforming existing solutions [38].

Another example of applying differential privacy in machine learning is the DP-SGD algorithm by Abadi et al. [39], used to achieve privacy for training data in a Convolutional Neural Network (CNN) model. CNNs are a type of neural network useful for pattern recognition and data classification. The DP-SGD algorithm minimizes the loss function of the CNN model by computing the average gradient of the loss function with added noise. The training set is divided into micro-batches to reduce memory consumption and improve computation performance. The gradient is clipped to control sensitivity and bias, then averaged and noised to protect training data privacy. The model parameters are updated using the learning rate, clipped gradient, and noise, ultimately returning optimal model parameters and quantifiable privacy loss [39].

Figure 2 shows a DP-SGD model on a CNN, where gradients are computed per training example, resulting in computational overhead [40].

### 3.2.2 Federated learning

A study proposed using federated learning (FL) to address challenges in sharing genomics data and handling large volumes of data [41]. FL trains a machine learning model using data from multiple devices without distributing the data, addressing privacy, security, and access control issues. Figure 3 shows devices training models locally and sending them to a centralized server to create a global model. FL generates valuable insights from sensitive data, but additional safety mechanisms are needed. Aziz et al. used differential privacy and the exponential mechanism to protect genetic data privacy, creating private histograms and applying differential privacy after data dimension reduction [41].

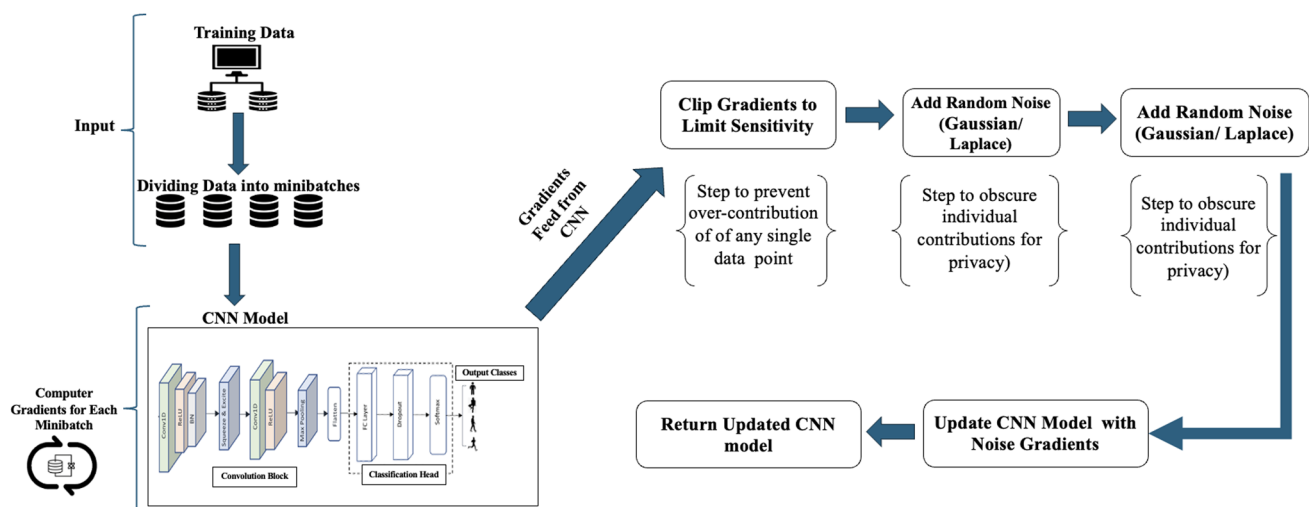
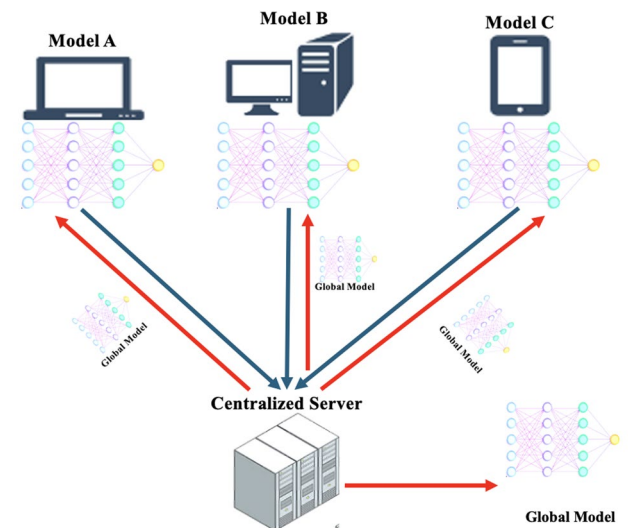


Fig. 2 DP-SGD on CNN model computing gradients on a per training example basis

Fig. 3 Federated Learning approach



Data-private collaborative learning (DPCL) enables model training across institutions without sharing patient data, unlike the Collaborative Data Sharing approach (CDS) [42]. Examples include Cyclic Institutional Incremental Learning (CIIL) and Institutional Incremental Learning (IIL), which are prone to “catastrophic forgetting” [43]. In IIL, each institution trains the model once before passing it to the next institution, while in CIIL, institutions can train more than once with a fixed number of training epochs [42]. Also, Sheller et al. showed that FL converges faster and performs better than CDS, CIIL, and IIL, with less bias due to aggregated results [42].

FL addresses privacy, data ownership, and technical concerns, encouraging multi-institutional collaborations and leading to accurate models for medical applications [42]. FL complies with HIPAA and GDPR but is vulnerable to privacy attacks like inference, reconstruction, and backdoors [44, 45]. Researchers should conduct privacy analysis and supplement models with other privacy-preserving techniques [46, 47].

### 3.3 Cryptographic solutions

To address concerns related to outsourcing genomic data to untrusted cloud environments, unauthorized use beyond agreed research scopes, and potential privacy breaches, cryptographic techniques have been developed to safeguard genomic data and prevent privacy leakage. One widely studied approach is Homomorphic Encryption (HE), which enables computations on encrypted data without requiring decryption. This technique allows secure outsourcing of statistical

operations, such as allele frequency calculations and regression modeling, to cloud platforms [3, 48, 49]. Mathematically, HE follows the principle:

$$\forall m_1, m_2 \in M, D(E(m_1) \circ_e E(m_2)) = m_1 \circ_p m_2 \quad (9)$$

where  $m_i$  represents a message,  $E$  and  $D$  are encryption and decryption schemes, and  $\circ_p$  and  $\circ_e$  are operations on plain and encrypted messages, respectively. HE schemes are classified as partially HE (supporting only addition or multiplication), somewhat HE (allowing limited operations), and fully HE (enabling arbitrary computations on encrypted data).

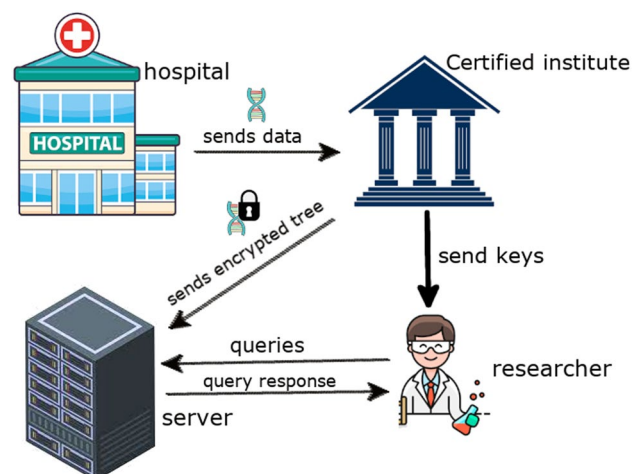
Kim et al. [50] proposed an HE-based framework to protect genotype data during imputation. This allows computations to be performed securely without exposing sensitive genetic sequences. Additionally, Mahdi et al. [24] improved privacy-preserving genomic searches by using an encrypted Generalized Suffix Tree (GST). In this method, genomic data is encrypted before storage. Secure access is ensured through a cryptographic protocol. This framework, as shown in Figure 4, involves multiple entities, including hospitals, research institutions, and cloud servers. These entities manage encryption keys and enable secure queries using Garbled Circuit (GC) protocols [51].

Another cryptographic approach is Secure Multi-Party Computation (SMC). This method allows multiple parties to analyze genomic datasets together without exposing individual data points [2]. In SMC, sensitive genomic data is split into *secret shares* and distributed among multiple entities. No single entity has access to the complete data. Computation is carried out over these shares using cryptographic protocols such as additive secret sharing, Shamir's secret sharing, or garbled circuits [52]. Each party performs operations only on its share of the data, and the intermediate results are combined at the end to reconstruct the final output. This approach ensures data privacy throughout the process. Many SMC implementations use homomorphic encryption and secret sharing to improve data security. GC-based techniques provide additional privacy protection [3, 53]. However, despite its privacy benefits, SMC has scalability issues. High computation and communication costs make it impractical for large-scale genomic datasets.

To overcome some of these limitations, researchers have explored Secure Cryptographic Hardware (SCH) solutions. One example is Intel Software Guard Extensions (SGX), which creates secure enclaves within CPUs to protect data during computation [54, 55]. Unlike software-based encryption techniques, SCH provides hardware-enforced security. This ensures that genomic data remains protected even during processing. Chen et al. [56] introduced an SCH-based technique to securely outsource genetic testing to untrusted cloud environments. This makes SCH a scalable alternative for genomic privacy. However, SCH still has challenges. It struggles to handle large-scale genomic datasets due to limited memory capacity. It also has potential security vulnerabilities that need further investigation [3].

Apart from cryptographic encryption, Differential Privacy (DP) has been widely used to enhance genomic privacy, especially in Genome-Wide Association Studies (GWAS). DP adds carefully calibrated noise to summary statistics, such as Minor Allele Frequencies (MAFs), chi-square ( $\chi^2$ ) statistics, and P-values. This prevents adversaries from inferring individual genomic data [9]. While DP helps reduce privacy risks, it often lowers data utility. To balance privacy and accuracy, researchers have explored relaxed DP models, such as Pufferfish Privacy [57], Bayesian Differential Privacy [58], and Rényi Differential Privacy (RDP) [59], which adjust adversarial assumptions or redefine sensitivity to improve usability while maintaining privacy as applicable to protecting against membership inference attacks [60].

**Fig. 4** System model for the proposed privacy-preserving string search on encrypted data



| Technique                      | Overheads   |               |         |               | Advantages                                                                                                                                             | Limitations                                     |
|--------------------------------|-------------|---------------|---------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
|                                | Computation | Communication | Storage | Accuracy Loss |                                                                                                                                                        |                                                 |
| Differential Privacy           | Medium      | Low           | Medium  | High          | Provides privacy for individuals                                                                                                                       | High accuracy losses, low data utility          |
| Homomorphic encryption         | High        | Medium        | High    | Medium        | Encrypted data can be computed, Provides semantic security, Input and output data privacy is secured                                                   | Computationally expensive, limited operations   |
| Secure multi-party computation | Medium      | High          | Medium  | Medium        | Private computation involving multiple parties                                                                                                         | High data transmission cost, scalability issues |
| Secure Cryptographic hardware  | Low         | Low           | Low     | Low           | Efficient computation and communication, few accuracy loss, support any mathematical operation, provide security against adversaries, ensure integrity | Memory size restrictions                        |

**Fig. 5** The comparison of cryptographic techniques and a data perturbation technique, which is made on overheads, benefits, and limitations of each technique

**Fig. 6** A depiction of a blockchain network that controls data access using multiple parties that hold shares of an encryption key. The blockchain stores immutable and transparent records, tracking consumers' written consent and researchers' requests to access data

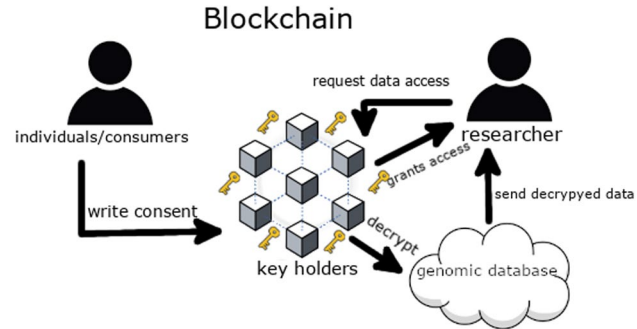


Figure 5 provides a comparative analysis of these cryptographic techniques, highlighting their respective strengths, limitations, and applicability in genomic privacy protection.

3.4 Blockchain solution

Blockchain is a shared, immutable database that records and distributes digital information among authorized participants [61]. It has been considered for genomic data sharing while protecting privacy [62, 63]. Researchers have proposed using blockchain to securely share GWAS datasets or machine learning (ML) model parameters trained on genomics data [64]. Figure 6 illustrates how blockchain functions in genomics. Access to genomic data is controlled by multiple independent parties holding shares of a split encryption key [16]. The blockchain ensures immutable storage of individuals' consent and data access requests [16]. Key features include user anonymity, access control, record auditability, and secure data analysis [62].

Grishin et al. [62] proposed pseudo-anonymous transactions using blockchain addresses for genetic testing. This method hides identity information and reduces the need for data de-identification. They also introduced delegated access control, where multiple independent organizations manage data access [62]. Blockchain further enables tamper-proof record-keeping by storing data access requests and user consent [62]. When an investigator requests access, encryption key holders read permissions from the blockchain, decrypt the data, and grant access if authorized [62]. This model supports secure computing by “bringing algorithms to the data,” reducing exposure of sensitive information. Grishin et al. [62] recommended privacy-preserving technologies like homomorphic encryption (HE) and secure multiparty computation (SMC) to enhance security. Recent advances have made these technologies more practical and scalable for direct-to-consumer (DTC) genomics companies [65].

**Table 2** Summary of privacy-preserving solutions for genomic data security

| Category                         | Techniques                                                   | Description                                                                                                       | Advantages                                                                                    | Limitations                                                                         | References      |
|----------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------|
| Data Perturbation Methods        | Differential Privacy (DP)                                    | Adds noise to genomic data before sharing or analysis to protect individual privacy.                              | Prevents re-identification and inference attacks; allows data sharing.                        | Reduces data accuracy; requires tuning to balance privacy and utility.              | [2, 23, 28, 29] |
|                                  | K-Anonymity Data Aggregation                                 | Generalizes or groups genomic attributes to prevent unique identification.                                        | Simple to implement; effective for small datasets.                                            | Vulnerable to linkage attacks; may not fully protect individual privacy.            | [26]            |
| Machine Learning-Based Solutions | DP-SGD (Differentially Private Stochastic Gradient Descent)  | Uses differentially private training methods to prevent data leakage in machine learning models.                  | Protects model training data from membership inference attacks.                               | Computational overhead; may degrade model performance.                              | [38–40]         |
|                                  | Federated Learning (FL)                                      | Enables decentralized model training on genomic data without sharing raw data.                                    | Reduces the need for data centralization; privacy-preserving AI training.                     | Vulnerable to inference and poisoning attacks; requires robust security mechanisms. | [41, 42, 41–42] |
| Cryptographic Solutions          | Homomorphic Encryption (HE)                                  | Allows computations on encrypted genomic data without needing decryption.                                         | Strong confidentiality; enables secure genomic cloud computing.                               | High computational cost; slow execution for large datasets.                         | [3, 48, 49, 80] |
|                                  | Secure Multi-Party Computation (SMC)                         | Enables collaborative genomic analysis while keeping individual data private.                                     | Ensures privacy even in multi-institutional studies.                                          | High communication and computation overhead.                                        | [2, 3, 53]      |
| Blockchain-Based Security        | Blockchain for Genomic Data Sharing                          | Uses distributed ledger technology to control access and maintain immutable records of genomic data transactions. | Enhances trust; ensures tamper-proof data sharing.                                            | High energy consumption; storage inefficiencies for large genomic datasets.         | [61–64]         |
|                                  | Generative Adversarial Networks (GANs) Synthetic Data Models | Uses deep learning to generate synthetic genomic datasets with privacy guarantees.                                | Allows privacy-preserving data sharing while maintaining statistical properties of real data. | May still be vulnerable to re-identification attacks if not carefully implemented.  | [66–68]         |
| Access Control Solutions         | Authority-Based Access Control                               | Restricts access to genomic data based on institutional approvals (e.g., NIH's controlled-access databases).      | Ensures oversight and policy enforcement for genomic data use.                                | Requires centralized control, potential misuse by authorities.                      | [69, 70]        |
|                                  | Participant-Based Access Control                             | Allows individuals to control who can access their genomic data.                                                  | Provides autonomy and user consent-based security.                                            | Relies on individuals' awareness and compliance, may limit research accessibility.  | [71]            |
|                                  | Trust-But-Verify Logging                                     | Enables auditing and tracking of genomic data access through immutable logs.                                      | Increases accountability and transparency in genomic data use.                                | Cannot prevent unauthorized access in real-time, only detects it after occurrence.  | [72, 73]        |

### 3.5 Synthetic data generation techniques

Researchers have proposed synthetic data generation to protect anonymity. Deep learning models like Generative Adversarial Networks (GANs) and Restricted Boltzmann Machines (RBMs) generate data that maintains the characteristics of the original data, reducing the risk of malicious inference [66]. However, synthetic data may not always guarantee privacy and utility. Oprisanu et al. [67] studied methods like Recombination model (Recomb), RBMs, GANs, WGANs, Rec-RBM, and Rec-GAN, measuring their utility and privacy. The Recomb model on the HapMap dataset had the best utility but reduced privacy. No single method was best across all metrics and datasets, indicating further research is needed to achieve high privacy and utility. Likewise, Jordan et al. in [68] used differential privacy in GANs with the PATE-GAN framework, bounding the influence of individual samples to ensure privacy and improve performance. Their experiments showed the synthetic data was as useful as the original and outperformed other methods.

### 3.6 Access control solutions

Preventing unauthorized access is crucial for genomics privacy and security. The U.S. National Institutes of Health (NIH) and the Wellcome Trust Foundation mitigated attacks by moving demographic information and GWAS summary statistics into controlled-access databases [69, 70]. Various access control models are used to prevent data breaches. Another model applicable in the context genomics privacy is using a trust-but-verify approach, recording and auditing transactions to detect malicious activities [71]. A participant-based model grants original participants control over their data and manages access through a trusted agent [71]. Furthermore, the NIH prefers an authority-based access control solution over identity-based methods. This context-sensitive model recognizes the source of authority, grants minimal necessary privileges, and uses the Research Authorization Service (RAS) to verify identities and issue authorization tokens [72]. However, NIH's reliance on commercial cloud storage, which uses identity-based solutions, poses security risks. More so, Mosca and colleagues highlight a significant concern indicating the genomic data protection through on controlled-access databases might not be adequately shielded from unauthorized users [73].

### 3.7 Translating privacy-preserving techniques into practice: real-world genomic frameworks

The Global Alliance for Genomics and Health (GA4GH) is a worldwide group that creates technical and policy guidelines to support the safe and ethical sharing of genomic data. It aims to balance data access with privacy protection. To achieve this, GA4GH has developed a set of standards that enable collaboration while keeping sensitive genomic information secure. These standards use data masking, encryption, and access control to reduce privacy risks in research. Some key standards include the Beacon Project, which allows controlled data queries, and the Passport Standard, which supports secure authentication across different systems. Other initiatives, like CanDIG, enable privacy-focused data analysis, while DUOS automates decisions on data use. These efforts show that privacy-preserving methods can work in large-scale genomic data management [74–77].

#### 3.7.1 GA4GH Beacon Project

The GA4GH Beacon Project helps researchers find genomic data while protecting privacy. It allows users to check if a specific allele is present. It gives a simple “yes” or “no” answer without revealing raw data. Beacons limit direct access to datasets and use query aggregation to lower the risk of identifying individuals. However, privacy risks still exist. If multiple Beacons are queried together, statistical methods could be used to infer sensitive information. To prevent this, stronger privacy techniques like differential privacy or threshold-based responses may be needed [74].

#### 3.7.2 GA4GH passport standard

The GA4GH Passport Standard introduces a federated authentication and authorization model. It grants role-based access to genomic datasets using machine-readable “visas.” These visas are individual assertions about a user's roles or data access rights. They define user permissions and are encoded within JSON Web Tokens (JWTs). Both Passports and Visas are digitally signed using cryptographic algorithms such as Secure Hash Algorithm (SHA), Rivest-Shamir-Adleman

(RSA), and Elliptic Curve (EC) [78, 79]. By using these cryptographic schemes, the GA4GH Passport Standard ensures that user permissions are securely transmitted and verifiable across different systems in the genomics and health data-sharing ecosystem. This approach improves interoperability and security. However, it still depends on centralized trust authorities, which could fail in multi-institutional collaborations. Also, verifying user credentials alone does not prevent unethical data use. This makes insider threat mitigation an ongoing challenge [75].

### 3.7.3 CanDIG federated data sharing

The CanDIG Federated Data Sharing platform is closely related to Secure Multi-Party Computation (SMC). It allows genomic analysis across institutions without revealing raw data (CanDIG Consortium, 2022). It ensures that institutions keep control over their datasets. This reduces the risks linked to centralized genomic repositories. However, SMC has major computational challenges. It requires high processing power and is complex to implement. This may limit its ability to handle large genomic studies. Federated querying helps protect raw data, but it does not fully eliminate the risk of inference attacks. Additional safeguards, like noise injection or differential privacy, are needed [76].

### 3.7.4 Data use ontology (DUO) system

The Data Use Ontology (DUO) system strengthens genomic privacy. It does this by enforcing policy-driven access restrictions (GA4GH Ethics and Policy Working Group, 2021). The DUO system employs structured Data Use Limitations (DULs) to automate access approvals. This ensures compliance with ethical and legal frameworks. This system reduces administrative burdens. However, it relies on predefined policies. This reliance may lead to inflexibility when addressing emerging privacy risks. A more adaptive governance model could enhance its effectiveness in preventing unauthorized access [77].

Overall, these genomic privacy frameworks show how privacy-preserving techniques are applied in genomic research. They prove the feasibility of federated authentication, cryptographic protections, and policy-driven oversight. However, challenges remain in scalability, computational efficiency, and resistance to inference attacks. Future advancements should focus on hybrid cryptographic methods and adaptive privacy-preserving techniques. These improvements will help strengthen genomic data security.

## 4 Laws and policies related to genetic information

### 4.1 Current policies

The rapid expansion of genomics and data-sharing technologies challenges genetic privacy regulation. Public policies aim to balance individual privacy with societal benefits, but defining privacy in the digital era remains complex [81]. Below are key regulations governing genetic data protection.

#### 4.1.1 Health insurance portability and accountability act (HIPAA)

HIPAA, enacted in 1996, regulates healthcare entities but excludes private companies [81]. Its privacy rule protects personal health data while allowing controlled use for public health [82]. The security rule mandates administrative, technical, and physical safeguards for electronic health information but excludes oral and written transmissions [82]. HIPAA permits data sharing without consent if de-identified using Safe Harbor or Expert Determination methods, which assess re-identification risks [83].

#### 4.1.2 General data protection regulation (GDPR)

The GDPR enforces strict privacy controls on personal data, including genomic information, processed by organizations handling EU citizens' data [84, 85]. It mandates data controllers to maintain security throughout data processing and ensures de-identified data remains irreversible [84]. Entities must adhere to seven principles: accountability, storage limitation, accuracy, data minimization, purpose limitation, integrity and confidentiality, and transparency [86].

### 4.1.3 Genetic information discrimination law and the genetic information privacy act

The Genetic Information Nondiscrimination Act (GINA) provides limited protection against genetic discrimination. It mainly applies to employment and health insurance [87, 88]. It does not cover epigenome, microbiome, or metagenome data [25]. Some states, like California, have stronger protections under the Genetic Information Privacy Act (GIPA). This law defines genetic data broadly and requires explicit consent for testing and data sharing [25, 89]. Other states, including Virginia and Colorado, have passed similar privacy laws [89–91]. Furthermore, GIPA requires direct-to-consumer (DTC) genetic testing companies to obtain consent before collecting, using, or sharing genetic data. Consumers have the right to revoke consent, delete data, and request account removal on platforms like *23andMe* [23]. Additionally, GIPA requires a court order for law enforcement or third-party access to genetic data [23].

### 4.2 Surreptitious genetic testing (SGT)

SGT refers to collecting and analyzing DNA without a person's knowledge or consent. This raises ethical and legal concerns. No federal law explicitly bans SGT, but some states have restrictions with different penalties [81, 92, 93]. GINA and the Affordable Care Act (ACA) provide limited protections. They apply only to employment and health insurance [81]. DTC companies have created internal policies that require users to verify ownership or get consent before submitting genetic samples. However, enforcement is inconsistent. A major case involving the Golden State Killer highlighted these concerns. Investigators used a public genealogy database (GEDMatch) and secretly collected DNA from a discarded tissue and a car door handle [81]. Police obtained a court order for genetic testing, but they searched GEDMatch without prior approval. This raised privacy concerns. Most jurisdictions do not require warrants for surreptitious DNA collection. As a result, individuals remain vulnerable to forensic genetic surveillance [94].

## 5 Addressing genomics data privacy and security issues

In this section, we summarize current challenges and provide recommendations for securing genomics data.

### 5.1 Challenging problems in genomic data privacy and security

The rapid growth of genomic data sharing has increased privacy and security risks. This has created a need for strong cryptographic and privacy-preserving techniques. Traditional methods like homomorphic encryption (HE), secure multi-party computation (SMC), and federated learning (FL) provide security guarantees. However, they also cause computational inefficiencies. AI-driven attacks and the risk of quantum decryption add more challenges. To address these issues, genomic privacy frameworks must be redesigned. They should include AI-resilient security mechanisms and quantum-resistant cryptographic methods.

#### 5.1.1 Trade-offs in cryptographic security and computational efficiency for genomic data protection

Genomic data encryption has always struggled to balance privacy, efficiency, and usability. Techniques like HE and SMC provide strong security. They allow computations on encrypted data without revealing sensitive information. However, these methods create high computational overhead, making them impractical for large-scale genomic analysis and real-time applications [2, 3, 95]. In contrast, lighter cryptographic methods like Secure Hybrid Rivest-Shamir-Adleman (SHRSA) and Elliptic Curve Cryptography (ECC) are more efficient. They are widely used for secure genomic data transmission [96, 97]. For example, Barmana and Saha in [97] proposed a DNA Encoded Elliptic Curve Cryptography System for IoT security. However, the security of these cryptographic methods depends on the difficulty of certain mathematical problems. RSA relies on integer factorization, while ECC is based on the discrete logarithm problem. These problems are too complex for classical computers but can be solved efficiently using Shor's algorithm on a quantum computer [98, 99]. This growing vulnerability has increased interest in quantum-resistant cryptographic approaches, which are discussed further in Section 5.2. Additionally, privacy-enhancing techniques like Differentiable Private Stochastic Gradient Descent (DP-SGD) and federated learning (FL), detailed in Section 3.2, have been proposed for genomic security [38, 39, 41]. However, these methods face new risks from AI-driven attacks. These attacks use machine learning models to infer sensitive genomic information.

### 5.1.2 AI-driven privacy attacks in genomic data security

As machine learning becomes more important in genomic research, AI-driven privacy attacks have become a major concern. These attacks were first studied in general machine learning settings. Later research showed that they also apply to genomic data privacy, especially in federated learning and precision medicine. The most notable attack types include Membership Inference Attacks (MIA), Model Inversion Attacks, and Adversarial Attacks.

- **Membership inference attacks (MIA):** Shokri et al. in [100] introduced membership inference attacks. They showed that an adversary could analyze a model's responses to check if a specific data point was in the training set. This attack was first studied in general machine learning models. Later, researchers applied MIAs to genomic datasets. In this case, an adversary may try to find out if a person's genomic sequence was part of a biomedical research cohort or a federated learning model [101]. This attack weakens anonymization efforts and creates a serious privacy risk for genetic data-sharing initiatives.
- **Model inversion attacks:** Fredrikson et al. in [102] demonstrated model inversion attacks. In these attacks, adversaries can reconstruct approximate features of individuals based on a model's output. Their original study focused on reconstructing facial images. However, later research showed that the same method also threatens genomic privacy. Adversaries can use a trained model's outputs to recreate approximate genomic features of individuals from the training dataset [102]. This attack is especially concerning in personalized medicine. Genomic predictions could be reversed to reveal sensitive genetic predispositions.
- **Adversarial attacks:** Carlini & Wagner in [103] studied adversarial attacks. In these attacks, attackers add tiny changes that trick machine learning models into making wrong predictions. Their research mainly focused on image recognition systems. However, the same idea applies to genomic models, especially those used for variant classification and disease prediction. Misclassification caused by adversarial perturbations in genomic sequences or biomedical imaging could lead to incorrect disease diagnoses or altered genomic interpretations. This could seriously impact precision medicine applications.

To address these threats, researchers have explored different privacy-preserving AI techniques to improve genomic data security. One widely used approach is Federated Learning with Differential Privacy (FL + DP). This method trains models across multiple institutions without centralizing data. It also adds noise to updates, reducing the risk of membership inference attacks [104, 105]. FL + DP balances privacy and computational efficiency, making it a preferred solution for genomic data protection. However, too much noise from differential privacy can lower model performance, making it less effective in some applications. Another important defense is adversarial robustness techniques. These methods train models to resist adversarial attacks and improve protection against data manipulation [19, 106, 107]. Despite their advantages, they require high computational power and may not work well across different genomic datasets. This makes large-scale implementation difficult. Homomorphic encryption (HE) has also been proposed for protecting model updates in federated learning. HE ensures that model parameters stay encrypted during transmission [101]. While HE prevents direct data exposure, it is computationally expensive. It also does not stop model inversion attacks, especially if an adversary gains access to the trained model [101]. These trade-offs highlight the need for hybrid security frameworks. Combining AI-specific defenses with cryptographic techniques can protect genomic privacy while keeping models usable and scalable.

### 5.2 The need for quantum-resistant cryptographic methods

As artificial intelligence-driven attacks create immediate privacy concerns, quantum computing introduces long-term risks to genomic data security. Classical encryption methods like RSA (Rivest-Shamir-Adleman) and ECC (Elliptic Curve Cryptography) rely on mathematical problems. Quantum computers can solve these problems much faster using Shor's algorithm [98, 99]. Since genomic data is often stored forever, future advancements in quantum computing could break existing encryption methods. This would leave sensitive genetic information vulnerable to decryption. To solve this problem, researchers have proposed post-quantum cryptographic (PQC) methods. These methods remain secure even against quantum-enabled attackers. Several PQC techniques have been explored for protecting genomic data, including:

- **Lattice-based cryptography:** Lattice-based encryption is a top choice for post-quantum security because its mathematical foundations resist quantum attacks [108]. This method relies on the difficulty of problems like Learning

with Errors (LWE) and Shortest Vector Problem (SVP). These problems are too complex for even quantum computers to solve. In genomic applications, lattice-based cryptography has been used in privacy-preserving genetic testing models. Researchers have developed homomorphic encryption schemes based on lattices. These allow secure computation on encrypted genomic data without revealing raw genetic sequences [108]. This is especially useful for genomic susceptibility testing, where privacy concerns prevent direct data sharing.

- **Hash-based signatures:** Hash-based cryptographic signatures offer a quantum-resistant way to ensure the integrity and authenticity of genomic data transactions. Unlike RSA or ECC signatures, which rely on integer factorization or discrete logarithms, hash-based signatures use cryptographic hash functions. This makes them naturally secure against quantum attacks. These methods are especially useful in genomic data-sharing frameworks. They help create tamper-proof audit logs, which are essential for complying with data protection regulations [109, 110].
- **Code-based encryption:** The McEliece cryptosystem is one of the oldest post-quantum cryptographic techniques. It is based on error-correcting codes and offers strong resistance to quantum decryption. Although it is not widely used in genomics yet, its ability to secure large-scale datasets makes it a possible option for genomic privacy frameworks in the future [111].
- **Blockchain-integrated based:** A recent approach combines Hyperledger Fabric, a permissioned blockchain platform, with post-quantum cryptography (PQC) to improve genomic and healthcare data security. This framework uses a distributed ledger system to ensure data integrity and confidentiality. It also helps overcome the challenges of centralized genomic data sharing. By integrating quantum-resistant cryptographic algorithms, the system protects encrypted genomic data from future quantum decryption threats. At the same time, it allows secure and monitored access. The permissioned blockchain structure enables controlled data sharing. This supports collaborative research while protecting individual privacy. Using PQC in blockchain-based genomic security frameworks is an important step toward long-term genomic data protection. It ensures that sensitive genetic information stays secure, even in the post-quantum era [112].

Despite the strong security of lattice-based, hash-based, blockchain-integrated, and code-based cryptography, using post-quantum cryptographic methods in genomic privacy frameworks is still a challenge. Current efforts focus on balancing computational efficiency with security. Some quantum-resistant methods need a lot of processing power, which makes real-time genomic analysis difficult. Future research should focus on adapting PQC techniques for genomic applications. These methods must meet the performance, scalability, and interoperability needs of modern genomic databases. As quantum computing technology improves, post-quantum cryptography will be essential for protecting genomic data from future decryption threats. This will help ensure long-term genomic privacy and security in precision medicine and biomedical research.

### 5.3 Integrating AI security and quantum cryptography into existing frameworks

To ensure long-term genomic privacy, security models must evolve to address both AI-driven threats and quantum decryption risks. Several genomic security frameworks, including those by Naveed et al. in [31], Ayday et al. in [113], and the Global Alliance for Genomics and Health (GA4GH) [74, 75], use policy-driven access control and encryption to protect genomic data. However, these frameworks mainly focus on access control and structured data sharing. They do not have specific measures for quantum-resistant cryptographic methods or AI-driven adversarial threats.

The Naveed et al. framework, shown in Figure 7, uses end-to-end encryption and controlled access to secure genomic data. It applies cryptographic data partitioning techniques to prevent large-scale data breaches [31]. While this is effective

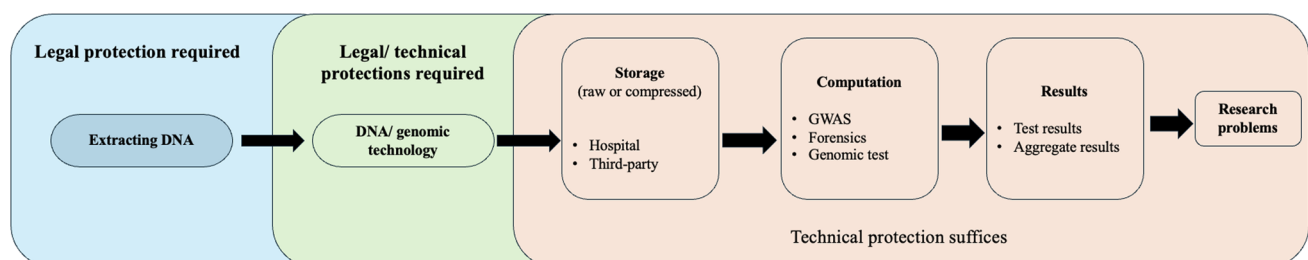


Fig. 7 The genomics data handling framework

against conventional threats, it does not include quantum-resistant encryption. It also lacks protections against AI-driven inference attacks like membership inference and model inversion attacks. Similarly, the GA4GH framework sets strong data-sharing standards. It includes the GA4GH Passport Standard for access control and Beacon and DUO (Data Use Ontology) for secure genomic data queries and automated governance [74, 75, 77]. Despite these strengths, the framework does not enforce post-quantum cryptographic (PQC) solutions. It also lacks strategies to defend against adversarial AI threats.

A notable encryption approach in genomic security is homomorphic encryption-based genotype and phenotype security (HEGP), proposed by Ayday et al. in [113]. HEGP allows secure computation on encrypted genomic and phenotypic data without revealing raw sequences. This makes it highly useful for privacy-preserving genomic research [113]. However, homomorphic encryption is not naturally quantum-resistant. Without adaptation for post-quantum security, it could be vulnerable to Shor's algorithm-based decryption attacks [98, 99].

To fix these issues, genomic security frameworks must include adversarially robust AI models, quantum-safe encryption, and policy-driven access control with post-quantum cryptography. AI models used in genomic research should be protected against adversarial attacks. Techniques like differentially private learning and adversarial training can help prevent membership inference and model inversion threats [114]. Additionally, genomic security must move from classical encryption to post-quantum cryptographic (PQC) methods. This includes lattice-based encryption, blockchain encryption, and hash-based digital signatures to defend against quantum-enabled attacks [112, 115]. Finally, policy-driven access control should be improved with quantum-resistant authentication. This will help ensure long-term genomic security against future quantum threats.

By adding AI-resilient defenses and quantum-resistant cryptographic techniques to frameworks like Naveed *et al.*'s and GA4GH standards, a hybrid genomic security model can be created to protect genomic privacy. This approach will reduce immediate AI threats, shield genomic data from future quantum decryption risks, and ensure long-term privacy in large-scale federated research. Genomic security frameworks must go beyond classical cryptography and structured access control. They need to evolve into AI-adaptive, quantum-secure privacy models. These models must be strong enough to handle both current and future threats in genomic research.

## 5.4 Final considerations and future directions

As genomic data privacy challenges continue to grow, a multi-layered approach is needed to process, store, and share genomic information securely. The use of cloud-based storage, federated learning, and blockchain technologies for genomic data analysis is increasing. However, these technologies also create serious security risks. Existing security frameworks provide important protections. However, new threats from adversarial AI attacks and the rise of quantum computing require stronger privacy-preserving methods.

This review highlights major progress in genomic data security. It covers cryptographic solutions, access control mechanisms, federated privacy frameworks, and policy-driven governance models. However, as discussed in Section 2, genomic databases are still highly vulnerable to re-identification attacks, especially through linkage and membership inference techniques. Access control mechanisms like the GA4GH Passport Standard help prevent unauthorized access. However, they rely on traditional cryptographic methods such as SHA, RSA, and elliptic curve, making them vulnerable to future quantum-based attacks (Section 3.7.2). Additionally, machine learning-driven genomic models introduce new risks. AI-enabled attacks can extract sensitive genomic attributes using model inversion and adversarial manipulation (Section 5.1.2). To solve these critical issues, future research should focus on the following key areas:

- **Strengthening AI-resilient privacy mechanisms:** AI-driven genomic models need stronger defense mechanisms. Techniques like differentially private learning and adversarial robustness can help prevent membership inference and model inversion attacks (Section 5.1.2). These defenses should be designed specifically for genomic data while ensuring good model performance and usability.
- **Advancing quantum-resistant cryptographic solutions:** As discussed in Section 5.2, classical encryption methods like RSA and ECC are vulnerable to Shor's algorithm. This makes it necessary to add post-quantum cryptographic (PQC) techniques to genomic security models. Lattice-based encryption, hash-based signatures, and code-based cryptographic methods must be assessed for their computational efficiency and compatibility with genomic data-sharing platforms.
- **Developing hybrid security frameworks for genomic privacy:** The combination of AI-specific privacy techniques and strong cryptographic security is essential. Future genomic security models should include policy-driven access

control, such as the GA4GH Passport Standard, along with quantum-safe encryption to ensure long-term data protection (Section 3.7.2). Additionally, privacy-preserving blockchain frameworks like Hyperledger Fabric with PQC should be further explored for secure data transactions and auditability (Section 5.2).

- **Ensuring scalability and efficiency in privacy-preserving techniques:** Many cryptographic techniques, such as fully homomorphic encryption (FHE) and secure multiparty computation (SMC), require high computational power. This makes them difficult to scale for large genomic datasets (Section 3.3). Future research should focus on improving these methods for real-world use while ensuring strong privacy protection.
- **Redefining regulatory and ethical standards for genomic data use:** Existing privacy laws and regulations, such as HIPAA, GDPR, and the GA4GH policy framework, must evolve to address the unique challenges of genomic data privacy (Section 4). Dynamic consent mechanisms should be used to give individuals more control over how their genetic data is used. This will help ensure compliance with ethical and legal guidelines.

By addressing both immediate risks from AI-driven attacks and long-term vulnerabilities introduced by quantum computing, genomic privacy frameworks can be adapted to meet future security demands. A multi-disciplinary approach combining cryptography, AI security, and legal oversight is necessary to ensure that genomic data remains secure while fostering innovation in precision medicine and biomedical research.

## 6 Conclusion

Advancements in genome sequencing and genetic analysis technologies have significantly reduced associated costs. This progress has driven increased accessibility and widespread data sharing. However, greater accessibility brings important security and privacy concerns. These concerns include vulnerabilities to persistent adversaries, sophisticated AI-driven threats like membership inference and model inversion attacks, and emerging quantum-enabled risks. Current privacy-preserving methods such as anonymization, differential privacy, homomorphic encryption, and secure multi-party computation offer valuable protections. Despite their benefits, these methods have significant limitations when facing advanced threats.

Future genomic privacy frameworks should incorporate quantum-resistant cryptographic approaches such as lattice-based encryption, hash-based signatures, and blockchain-integrated cryptography. It is also crucial to enhance AI model resilience by using robust defenses like differentially private training and adversarial robustness methods. Additionally, addressing scalability and computational efficiency is essential. Techniques like blockchain-integrated cryptography and Secure Cryptographic Hardware (e.g., Intel Software Guard Extensions-SGX) are promising solutions for reducing computational overhead. Elliptic Curve Cryptography (ECC) remains a useful interim solution, despite its future vulnerability to quantum computing.

Global genomic data sharing is complicated by diverse regulatory frameworks. Regulations such as GDPR in Europe, HIPAA in the United States, and state-specific laws like California's GIPA create complex compliance requirements. These differences often require dual compliance strategies, limiting effective international collaboration. Transparent and informed consent processes remain essential for ethical practices and user trust. Real-world frameworks like GA4GH Beacon and Passport standards demonstrate current efforts but highlight areas that require further development.

To sum up, addressing genomic data privacy requires a comprehensive, multidisciplinary approach. This approach should integrate robust technical defenses, harmonized regulatory standards, and clear ethical guidelines. Implementing these integrated solutions will ensure genomic data security keeps pace with evolving threats. Ultimately, this fosters responsible and innovative progress in precision medicine and genomic research.

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